Problem: is the viral “gene package” neutral?

Giant viruses (NCLDV) carry many genes stolen from hosts

Host alga
Chlorella proteome

HGT →

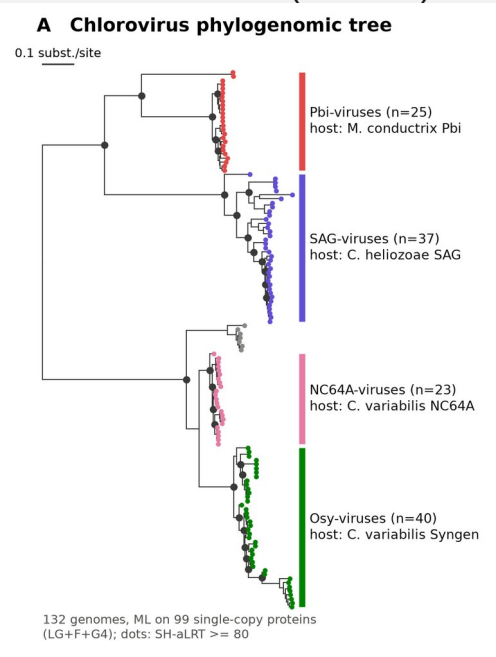
Virus genome, ~330 kb
~7 % of genes: algal-like accessory

After the transfer, what does evolution do with the gene?

H0: neutral cargo
- dN/dS (ω) ≈ 1
- gene copies decay
- promoter drifts like synonymous sites

H1: domesticated
- ω < 1 (purifying)
- conserved blocks in the inherited host domain
- constrained promoter

Model: chloroviruses (4 clades)



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Van Etten et al. 2020 Viruses; Moniruzzaman et al. 2020 Nat Commun; Irwin et al. 2022 Nat Microbiol

Methods: phylogenomics + per-gene selection tests

Data: 132 chlorovirus genomes (NCBI, 20 Oct 2026); proteomes of C. variabilis + 3 green algae

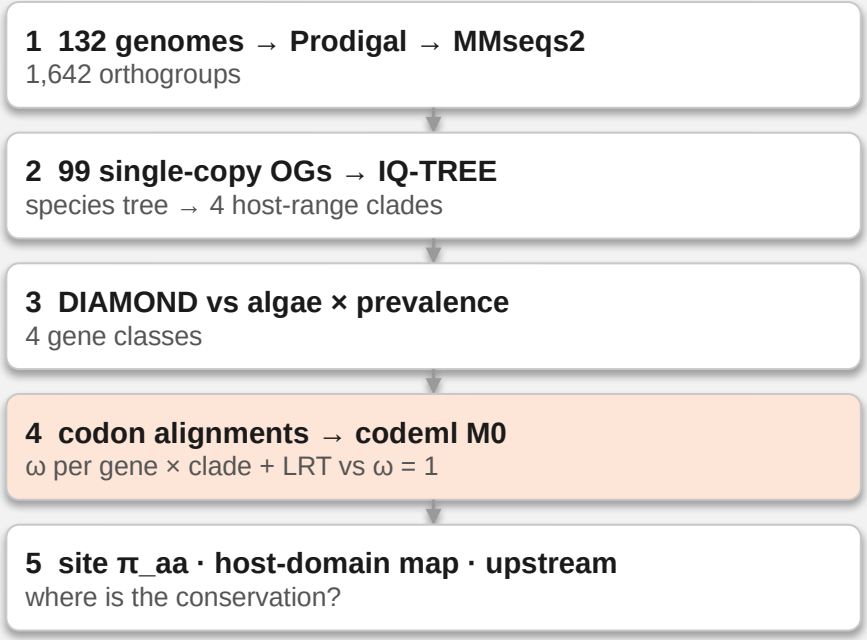
Classes: core = OG in ≥ 90 % genomes; algal-like = DIAMOND hit (e ≤ 1e-5) in ≥ 50 % of members

Stolen gene package = accessory + algal-like (144 OGs, 7 % of genes)

Selection: codeml M0 per gene × clade, LRT vs ω = 1 (BH); filter 0.01 < dS < 1.5

Stats: Wilcoxon, Kruskal–Wallis, Mann–Whitney + Cliff's δ, OLS (clade, dS, length; SE clustered by OG)

Pipeline (Python + bioconda; run_all.sh)



Prodigal · MMseqs2 · DIAMOND · MAFFT · IQ-TREE 3 · PAML 4.10

Results 1: stolen genes are under purifying selection

97 %

of stolen genes reject ω = 1 (114/117, LRT q < 0.05)

0.10

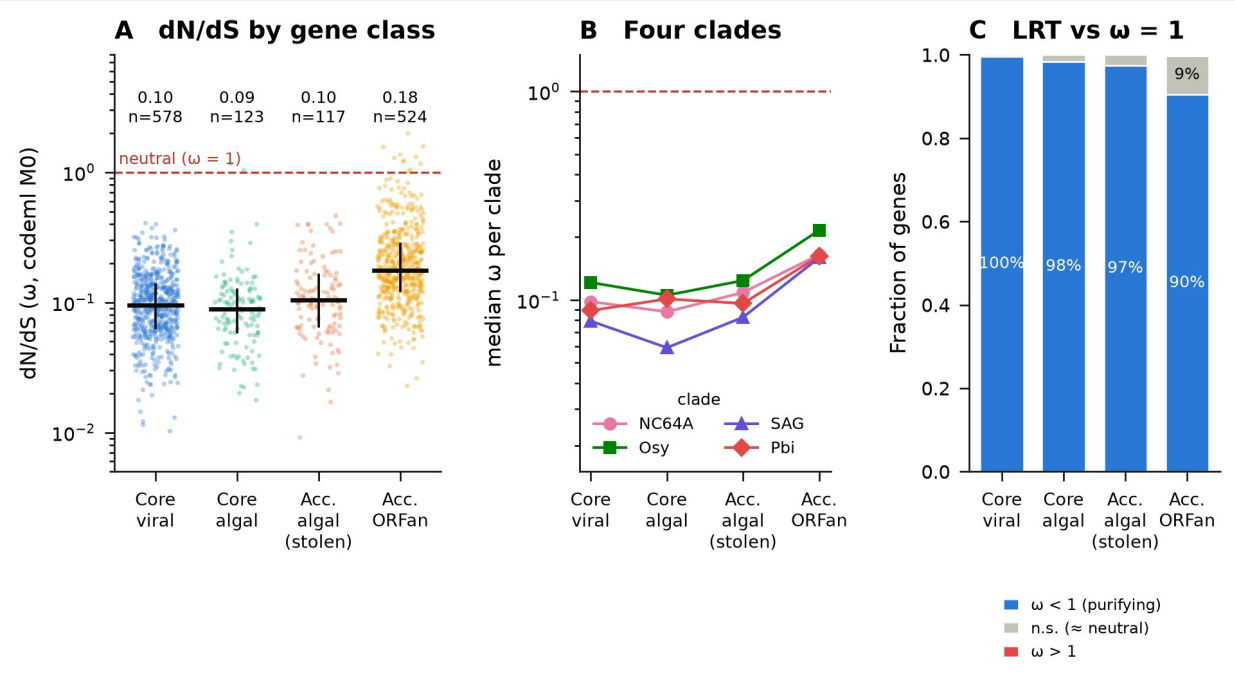
median ω, same as core viral genes (0.095); OLS β = +0.04, p = 0.34

ORFans are the loosest: ω = 0.18, 9 % not different from neutral

Same in all 4 clades; ω does not depend on identity to the host homolog (ρ = −0.10, p = 0.36)

→ **H0 (neutral cargo) is rejected**

Fig. 2. dN/dS per gene × clade (n = 1,342). A: by class; B: median per clade; C: per-gene LRT against ω = 1



codeml M0 (PAML); BH correction; all numbers in results/stats.txt

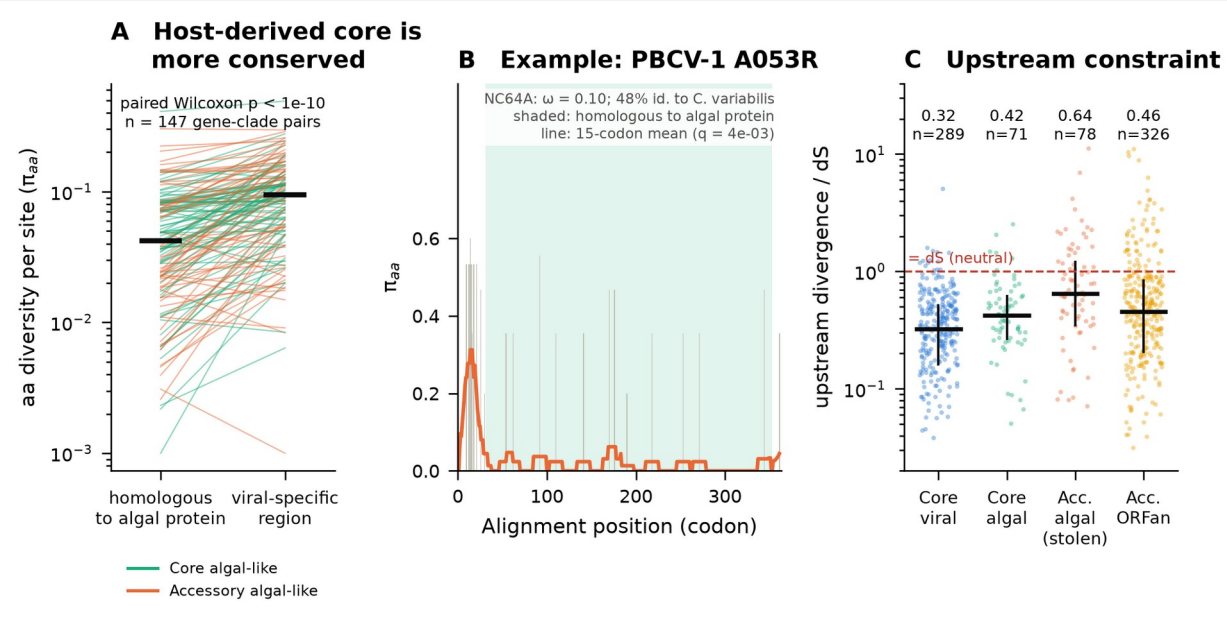
Results 2: the host domain is kept, the promoter is looser

Host domain conserved: π_aa 0.031 vs 0.108 in viral additions (p = 1e-10; 84 % of stolen genes)

Conserved blocks in 51 % of stolen vs 38 % of core genes (χ^2 p = 0.03)

Promoter constrained but looser: d_up/dS 0.64 vs 0.32 in core genes (p = 8e-9, δ = 0.43)

Fig. 3. A: aa diversity in host-homologous vs viral-specific regions; B: example gene PBCV-1 A053R; C: upstream divergence / dS



Conclusion

The stolen package is domesticated: the virus keeps the host function, while its regulation is still being wired in. The truly neutral candidates are the ORFans.

Limits: algal homology ≠ proven host origin; one ω per gene (M0); promoters only where intergenic ≥ 40 nt